



Netflix with HDR

Netflix has just published HDR content on their website starting with Marco Polo.
<http://dgit.in/NetflixHDR>



Uber for auto-rickshaws

Jugnoo just roped up \$10 million funding in series B for the startup.
<http://dgit.in/jugnoo10m>



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A Guide to Audio connectors and standards

Finding compatible headsets can be a challenge owing to various audio connectors and standards out there. So let this guide help you out!



Sheetal Rao

Last week I was on a hunting spree. The aim was to find the perfect IEM, that could provide decent audio output and fit in my budget as well. I have an iPod (whose headphone have gone missing) as well as an Android smartphone. How difficult would it be to find an IEM that would work with both, quite difficult, apparently. None of them worked properly with both my devices. There

would be an issue with either the remote controls on the IEM or with the microphone. Why should the phone manufacturer decide which headphones I buy even when all of them come with a 3.5mm connector? That was the call, I had to know, WHY!

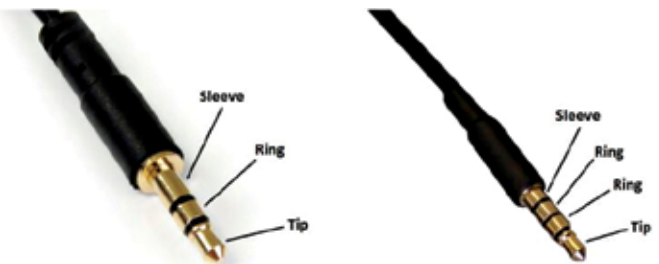
The old school portable music players, like the iPods or MP3 players had a single 3.5mm 3-conductor jack that later went on to become universal. It came into prominence with the first Sony Walkman, which was introduced in 1979. Life was much more simpler before the smartphone era, at least in the audio headset department. A phone audio connector, also known as the phone jack or audio jack is commonly used for analog signals, primarily audio. As our phones and other music devices evolved, so did these plugs and connectors. In due course, many audio connector standards came up and left it to the manufacturers to decide the standard they wanted to adopt. Even though all these plugs are cylindrical in shape, the bands on the plug define the number of contact points and their arrangements.

Understanding TS, TRS and TRRS

Audio connectors come in various sizes, most common ones being the 2.5mm, 3.5mm and 6.35 mm ones. It's easier and more technically accurate to refer to each by their Tip-Ring-Sleeve configuration. The Tip carries the signal and the sleeve or the body provides grounding. The very earlier connectors comprised of the carrying mono sound via the tip and the body i.e. the Sleeve provided grounding, and thus were called the TR connectors.

Later on, rings could be seen added onto the entire arrangement, giving birth to the TRS configuration where rings would be added between the Tip and the Sleeve. The one's with single rings were referred as TRS and dual ones as TRRS. Introduction of rings in the entire assembly made it possible to interchange the

functions between the Tip, the Rings and the Sleeve. For example, in a TRRS arrangement, we could have the Tip carrying the left audio, the first Ring carrying the right audio, the second Ring carrying the microphone signals and the Sleeve would provide grounding or the Sleeve would carry microphone signals and the second Ring would provide grounding.



TRS Audio Plug

TRRS Audio Plug

TRRS Standards

2.5mm and 3.5mm sockets with TRS and TRRS provided mono or stereo sound, a microphone input, and the signaling. The 3-conductor 2.5 mm connectors were on old feature phones, while the four-conductor 3.5 mm connectors were adopted by the newer smartphones. There was no recognised standard that the manufacturers would follow while using the TRRS connectors with three conductor TRS that could be used uniformly.

The four conductors of a TRRS connectors are assigned to different purposes by different manufacturers. Any 3.5 mm plug can be plugged mechanically into any 3.5mm socket, but combinations can be electrically incompatible. Plugging TRRS headphones into a TRS socket or vice versa or plugging TRRS headphones from one manufacturer into a TRRS socket from another may not allow the headphone to function correctly, or at all. Mono audio should always work, but stereo audio and microphone may not, depending on the wiring of the contacts on the socket as well as the connector.

Mainly, There are two different forms of format followed, namely OMTP and CTIA. OMTP stands for Open Mobile Terminal Platform whereas CTIA for Cellular Telephone Industries